

Optimal carbon taxation with uninsurable idiosyncratic risk: preliminary draft

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Abstract

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1 Introduction

How should governments tax carbon in the presence of household inequality and risk? How does carbon taxation affect the provision of insurance, redistribution, and how does it impact welfare? In this paper, we develop a fiscal climate-economy model with incomplete markets to answer these questions.

To gain insight into our main quantitative results, we begin by analytically exploring the effect of inequality, risk, and borrowing constraints on the optimal taxation of carbon in a simple two-period model. We then present our main framework, which extends the climate-economy model of [Douenne et al. \(2023\)](#) featuring fiscal policy and heterogeneous agents by adding uninsurable idiosyncratic income risk as in [Aiyagari \(1994\)](#). While doing this prevents us from obtaining analytical solutions, we use advanced numerical methods to quantitatively study a Ramsey problem in this incomplete markets economy.

Our economy features a continuum of households with preferences over consumption, leisure, and the climate. Households' productivity is subject to idiosyncratic risk against which they cannot insure: they can only invest in a risk-free asset subject to a borrowing constraint. The final consumption-investment good is produced using capital, labor, and energy. Energy is produced in a second sector using capital and labor. The production of energy generates CO₂ emissions that firms can mitigate by paying abatement costs. Non-abated CO₂ emissions accumulate and cause climate change, that affects households through both production and direct utility damages. Since firms' decision to engage in abatement activities depends on the cost of CO₂ emissions, the government can mitigate climate change by setting a tax on emissions. In addition, the government has access to linear taxes on capital, labor, energy production, and lump-sum transfers, and uses these instruments to provide a public good, redistribution, and insurance against idiosyncratic productivity risk.

We calibrate our model to the U.S. economy. [...] Our approach is based on the global optimization algorithm of [Dyrda and Pedroni \(2023\)](#). [Quantitative approach: To be completed].

[Preview of the results: To be completed].

Related literature. Our paper contributes to the literature analyzing fiscal policy in the presence of environmental externalities. In particular, it contributes to the literature studying optimal carbon taxation in second-best economies and carbon taxation in the presence of risk and inequality.

An extensive literature studies optimal environmental taxes in the presence of distortionary taxation in static representative-agent frameworks (e.g., [Sandmo, 1975](#); [Bovenberg and de Mooij, 1994](#); [Bovenberg and van der Ploeg, 1994](#); [Bovenberg and Goulder, 1996](#)), later extended to a dynamic environment ([Barrage, 2020](#)). The main takeaway of this literature is that, in the presence of tax distortions, the optimal tax on carbon is different from—typically below—the social cost of carbon (SCC). Another stream of papers study optimal pollution taxation with distortionary taxes and heterogeneous agents (e.g., [Kaplow, 2012](#); [Jacobs and de Mooij, 2015](#); [Jacobs and van der Ploeg, 2019](#)). When tax distortions arise as an optimal response to inequality, it is in general no longer optimal to deviate from the Pigouvian principle, i.e. tax distortions do not justify taxing carbon below its social cost. [Douenne et al.](#)

(2023) generalize this result to a dynamic environment: in a climate-economy model based on Barrage (2020), they theoretically show that optimal carbon taxes are on average Pigouvian, and they find that temporary deviations from the SCC are quantitatively negligible. They also find that inequalities reduce the SCC by making consumption relatively more valuable, even though the effect does not appear to be quantitatively large. We contribute to this literature by quantifying optimal carbon taxes in an economy featuring uninsurable idiosyncratic income risk. Our approach also allows us to... [...] Compared to Douenne et al. (2023) who abstract from risk, the presence of incomplete markets provides a rationale for capital taxation, and we study how capital and carbon taxes interact. [To be completed].

An abundant literature also studies the distributional effects of carbon taxation (for recent examples, see van der Ploeg et al., 2022; Känzig, 2023). Closer to our work, several papers have recently studied this question within heterogeneous agents incomplete markets models. Fried et al. (2018) study the distributional effects of exogenous carbon tax reforms in an OLG economy with heterogeneous households exposed to idiosyncratic productivity shocks. In a follow-up paper, Fried et al. (2021) search for the optimal revenue-recycling to an exogenous carbon tax introduced in a sub-optimal tax system. Their approach abstracts from the climate-economy interaction and focuses on the steady-state. In contrast, we allow the planner to choose its policy instruments to maximize social welfare, which enables us to study the optimal carbon tax and how it interacts with other policies at the optimum. Our approach is also well suited to study distributional effects during the transition, and our climate model allows us to perform a comprehensive welfare analysis of climate policies. Another recent contribution to this literature is Benmir and Roman (2022) who study the distributional effects of implementing the net-zero target in the U.S. based on a HANK framework. Their economy accounts for multiple frictions relevant at the aggregate level, but they only consider exogenous policies.

Finally, our paper relates to the literature using advanced numerical methods to study carbon taxation in the presence of risk. A noteworthy example is Cai and Lontzek (2019) who study how the SCC is impacted by climate and (aggregate) economic risk. They use value function iteration to deal with the high dimensionality of their problem... [To be completed].

The remainder of the paper is structured as follows. Section 2 presents a simple two-period model to illustrate the main mechanisms at play. Section 3 lays out the infinite-horizon model, the planning problem, and the solution method. Section 4 details our calibration. Section 5 presents our main quantitative results. Finally, Section 6 concludes.

2 Two-period model

To illustrate the mechanisms driving the results of our infinite-horizon model, we first present a simple two-agent two-period example. To keep things as simple as possible, we consider a partial equilibrium endowment economy, and we abstract from modeling the environmental externality. Our simple model is based on the idea that the cost of carbon taxation is forgone consumption, and the benefits are the provision of a public good (from direct utility benefits) and higher future endowments (from the mitigation of production damages).

[To be completed: choose the relevant mechanisms to highlight].

3 Infinite-horizon model

In this section we set-up the infinite-horizon model where households face uninsurable idiosyncratic income risk and where the government aims to provide redistribution, insurance, and to address climate change. The presentation follows [Dyrda and Pedroni \(2023\)](#) to which we add an energy sector, climate change, and carbon taxation as in [Barrage \(2020\)](#) and [Douenne et al. \(2023\)](#).

3.1 Households

Time is discrete and the horizon is infinite. There is a continuum of households with preferences over consumption c_t , labor h_t , and the climate Z_t ,

$$\mathbb{E}_0 \left[\sum_t \beta^t u(c_t, h_t, Z_t) \right]. \quad (1)$$

Households' labor productivity is denoted by $e \in E$ with $E \equiv \{e_1, \dots, e_L\}$, and follows a Markov process with transition matrix Γ . The only asset available to households is a risk-free one, denoted a , which can take values in $A \equiv [\underline{a}, \infty)$. Thus, households are indexed by $(a, e) \in S$, with $S \equiv A \times E$.

Given a sequence of taxes on labor and capital income, $\{\tau_t^h, \tau_t^k\}_{t=0}^\infty$, transfers $\{T_t\}_{t=0}^\infty$, and prices $\{r_t, w_t, R_t\}_{t=0}^\infty$, with $R_t \equiv 1 + (1 - \tau_t^k)(r_t - \delta)$, in each period t each household chooses how much to consume, $c_t(a, e)$, work $h_t(a, e)$, and save, $a_{t+1}(a, e)$, to solve

$$v_t(a, e) = \max_{c_t, h_t, a_{t+1}} u(c_t(a, e), h_t(a, e), Z_t) + \beta \sum_{e_{t+1} \in E} v_{t+1}(a_{t+1}(a, e), e_{t+1}) \Gamma_{e, e_{t+1}} \quad (2)$$

subject to

$$(1 + \tau^c) c_t(a, e) + a_{t+1}(a, e) = (1 - \tau_t^h) w_t e h_t(a, e) + R_t a_t + T_t, \quad (3)$$

$$a_{t+1}(a, e) \geq \underline{a}, \quad (4)$$

where τ^c is an exogenous and constant consumption tax.¹

We denote aggregate consumption and hours worked with capital letters,

$$C_t = \int_S c_t(a, e) d\lambda_t, \quad H_t = \int_S h_t(a, e) d\lambda_t, \quad (5)$$

with $\{\lambda_t\}_{t=0}^\infty$ a sequence of probability measures defined over the Borel sets $\mathcal{S}$ of the space S , where the initial measure λ_0 is given.

3.2 Firms

There are two production sectors, each represented by a representative firm.

¹As in [Dyrda and Pedroni \(2023\)](#), we add this as a parameter for calibration purposes, but this is not an instrument for the planner.

3.2.1 Final good sector

The first sector produces the final consumption-investment good, Y_t , using a constant-returns-to-scale technology, $F(\cdot)$, with capital, labor, and energy inputs, denoted $K_{1,t}$, $H_{1,t}$, and E_t . Final good production is subject to climate damages $D(Z_t)$, such that

$$Y_{1,t} = (1 - D(Z_t)) A_{1,t} F(K_{1,t}, H_{1,t}, E_t), \quad (6)$$

with $A_{1,t}$ the total factor productivity of the final good sector. The representative final good firm chooses capital, labor, and energy given the respective real factor prices r_t , w_t , and p_t^e , and makes zero profit. The first order conditions are

$$r_t = (1 - D(Z_t)) A_{1,t} F_{K,t}, \quad (7)$$

$$w_t = (1 - D(Z_t)) A_{1,t} F_{H,t}, \quad (8)$$

$$p_t^e = (1 - D(Z_t)) A_{1,t} F_{E,t}. \quad (9)$$

3.2.2 Energy sector

The second sector produces energy, E_t , using a constant-returns-to-scale technology, $G(\cdot)$, with capital and labor inputs, $K_{2,t}$ and $H_{2,t}$, both assumed to be fully mobile across sectors. Energy production is given by

$$E_t = A_{2,t} G(K_{2,t}, H_{2,t}), \quad (10)$$

with $A_{2,t}$ the total factor productivity of the energy sector. Energy production is polluting, with industrial CO₂ emissions given by

$$E_t^M = (1 - \mu_t) E_t, \quad (11)$$

where μ_t represents the fraction of energy coming from clean technologies. The cost of emission abatement is given by $\Theta_t(\mu_t) E_t$, with $\Theta_{\mu,t}, \Theta_{\mu\mu,t} > 0$, and $\Theta_t(0) = 0$. The representative firm's profits in the energy sector are given by

$$\mathcal{P}_t = (p_t^e - \tau_t^i) E_t - \tau_t^e (1 - \mu_t) E_t - w_t H_{2,t} - r_t K_{2,t} - \Theta_t(\mu_t, E_t), \quad (12)$$

with τ_t^i the excise intermediate-goods tax on total energy production, E_t , and τ_t^e the excise tax on carbon emissions, E_t^M . Since the abatement cost function is linear in E_t , profits in the energy sector are null. The representative energy firm chooses capital, labor, and abatement such that

$$r_t = (p_t^e - \tau_t^i - \tau_t^e (1 - \mu_t) - \Theta_{E,t}) A_{2,t} G_{K,t}, \quad (13)$$

$$w_t = (p_t^e - \tau_t^i - \tau_t^e (1 - \mu_t) - \Theta_{E,t}) A_{2,t} G_{H,t}, \quad (14)$$

$$\tau_t^e = \frac{\Theta_{\mu,t}}{E_t}. \quad (15)$$

3.3 Government

The government has access to proportional income taxes on capital, τ_t^k , and labor, τ_t^h , taxes on total energy production, τ_t^i , on carbon emissions, τ_t^e , as well as a fixed consumption tax, τ^c . Each period it uses these instruments to finance an exogenous stream of expenses, G_t , and lump-sum transfers, T_t . The government can also issue debt, B_{t+1} , whose sequence must remain bounded. The government's inter-temporal budget constraint is

$$G_t + T_t + R_t B_t = \tau^c C_t + \tau_t^h w_t H_t + \tau_t^k (r_t - \delta) K_t + \tau_t^i E_t + \tau_t^e E_t^M + B_{t+1}. \quad (16)$$

3.4 Climate

We use the climate model of [Dietz and Venmans \(2019\)](#) in order to capture two key features of climate dynamics: the temperature response to emissions is almost immediate and permanent, and temperature is almost linear in cumulative emissions.² This model therefore correctly approximates the impulse response of temperature to emissions, which is essential for a proper quantitative assessment of the effect of climate policies on welfare. Formally, global mean surface temperature change relative to pre-industrial levels, Z_t , follows the law of motion

$$Z_{t+1} = Z_t + \epsilon(\zeta \mathcal{E}_t - Z_t), \quad (17)$$

with ζ the transient climate response to cumulative carbon emissions (TCRE, see [Dietz and Venmans, 2019](#)), ϵ a parameter for the speed of adjustment of temperature to an emission pulse, and $\mathcal{E}_t$ the cumulative emissions that evolve as follows:

$$\mathcal{E}_{t+1} = \mathcal{E}_t + E_t^M + E_t^{\text{ex}}, \quad (18)$$

where E_t^{ex} represents exogenous emissions (from land, or from the rest of the world).

3.5 Competitive equilibrium

Definition 1 Given K_0 , B_0 , an initial distribution λ_0 , and a policy $\pi \equiv \{\tau_t^h, \tau_t^k, \tau_t^i, \tau_t^e, T_t\}_{t=0}^\infty$, a **competitive equilibrium** is a sequence of value functions $\{v_t\}_{t=0}^\infty$, an allocation $X \equiv \{c_t, h_t, a_{t+1}, Z_t, E_t, \mu_t, K_{1,t}, K_{2,t}, K_{t+1}, H_{1,t}, H_{2,t}, H_t, B_{t+1}\}_{t=0}^\infty$, a price system $P \equiv \{R_t, w_t, r_t, p_t^e\}_{t=0}^\infty$, and a sequence of distributions $\{\lambda_t\}_{t=0}^\infty$, such that for all t :

1. the allocations solve the consumers and the firms problems given prices and policies;
2. the sequence of probability measures $\{\lambda_t\}_{t=1}^\infty$ satisfies

$$\lambda_{t+1}(\mathcal{S}) = \int_{\mathcal{S}} Q_t((a, e), \mathcal{S}) d\lambda_t, \quad \forall \mathcal{S} \text{ in the Borel } \sigma\text{-algebra of } \mathcal{S}, \quad (19)$$

²For a discussion of how these properties feature in other climate-economy models, see [Mattauch et al. \(2020\)](#) and [Dietz et al. \(2021\)](#). For further references on the temperature response to emissions over time, see [Joos et al. \(2013\)](#), [Ricke and Caldeira \(2014\)](#) and references therein. For further references on the linear relationship between cumulative emissions and temperatures, see [Matthews et al. \(2009\)](#), [Gillett et al. \(2013\)](#), or the summaries provided in [IPCC \(2021\)](#).

where Q_t is the transition probability measure;

3. the government budget constraint (16) is satisfied in every period, and debt is bounded;
4. temperature change satisfies equation (17) in every period, and;
5. markets clear, i.e., the following equations are satisfied:

$$H_t = H_{1,t} + H_{2,t}, \quad (20)$$

$$K_t = K_{1,t} + K_{2,t}, \quad (21)$$

$$C_t + G_t + K_{t+1} + \Theta_t(\mu_t, E_t) = (1 - D(Z_t))A_{1,t}F(K_{1,t}, H_{1,t}, E_t) + (1 - \delta)K_t, \quad (22)$$

$$E_t = A_{2,t}G(K_{2,t}, H_{2,t}), \quad (23)$$

$$H_t = \int_S e h_t(a, e) d\lambda_t, \quad (24)$$

$$K_t + B_t = \int_S a d\lambda_t. \quad (25)$$

3.6 Ramsey problem

We assume that the government announces and commits to a sequence of policies at time zero.

Definition 2 Given K_0 , B_0 , and λ_0 , for every policy π , **equilibrium allocation rules** $X(\pi)$ and **equilibrium price rules** $P(\pi)$ are such that $\{\pi, X(\pi), P(\pi)\}$ together with the corresponding $\{v_t\}_{t=0}^\infty$ and $\{\lambda_t\}_{t=1}^\infty$ constitute a competitive equilibrium. Given a welfare function $\mathcal{W}(\pi)$, the **Ramsey problem** is to $\max_{\pi \in \Pi} \mathcal{W}(\pi)$ subject to $X(\pi)$ and $P(\pi)$ being equilibrium allocation and price rules, and Π is the set of policies $\pi \equiv \{\tau_t^h, \tau_t^k, \tau_t^i, \tau_t^e, T_t\}_{t=0}^\infty$ for which an equilibrium exists.

If we assume that the planner has utilitarian preferences, then the planner's objective is given by

$$\mathcal{W}(\pi) = \int_S \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \tilde{\beta}^t u(c_t(a_0, e_0|\pi), h_t(a_0, e_0|\pi)) \right] d\lambda_0, \quad (26)$$

where $\tilde{\beta} \geq \beta$ represents the discount rate of the planner. Specifically, we follow [Farhi and Werning \(2007\)](#) and allow the planner to value the future more than households in order to reconcile the impatience of individual decision-makers with the more ethical approach to intertemporal welfare of the social planner.³

3.7 Solution method

The solution method builds on [Dyrda and Pedroni \(2023\)](#) and is adapted to our quantitative exercises, whose aim are to characterize i) how the presence of risk and inequality affect the optimal carbon tax,

³This disagreement over the discount rate can be microfounded by an OLG economy where individuals attach some altruistic weight to their offsprings, which the planner accounts for in addition to valuing future generations directly (see [Bernheim, 1989](#); [Farhi and Werning, 2007](#)). For applications of this modeling to climate change economics, see e.g. [Belfiori \(2017\)](#), [Barrage \(2018\)](#), and [van der Ploeg and Rezai \(2021\)](#).

and ii) how introducing carbon taxation affects the economy, and in particular how it affects risk and inequality.

[To be completed].

4 Calibration

This section describes the details of our calibration. All parameters are summarized in Table I of Appendix A.

Our calibration is based on the U.S. economy. We follow [Douenne et al. \(2023\)](#) and scale up the U.S. so that its emissions and GDP are those of the world, with population being adjusted to preserve the U.S. GDP per capita. A close alternative would be to simply consider the U.S. economy and assume that emissions from the rest of the world evolve proportionally to domestic ones. The caveat of this alternative approach is that it would lead the U.S. to ignore the negative impact of its emissions abroad, and consider only the U.S. SCC.⁴ Our approach is intended to ensure that policies reflect the global SCC, which is consistent with the stated objectives of the U.S. administration which claims that “It is essential that agencies capture the full costs of greenhouse gas emissions as accurately as possible, including by taking global damages into account.” (Executive Order 13990 of Jan 20, 2021).

4.1 Climate model

We calibrate the climate model of [Dietz and Venmans \(2019\)](#) based on [IPCC \(2021\)](#). We set the initial cumulative carbon emissions to $\mathcal{E}_{2020} = 2390\text{GtCO}_2$ and the initial temperature change to $Z_{2020} = 1.07^\circ\text{C}$. We take the report’s best estimate for the TCRE, at $\zeta = 0.00045^\circ\text{C/GtCO}_2$. For the speed of adjustment of temperature to an emission pulse, we follow [Dietz and Venmans \(2019\)](#) and set $\epsilon = 0.5$.

We calibrate initial industrial and land emissions from the Global Carbon Project ([Friedlingstein et al., 2022](#)), at $E_{2020}^M = 36.33\text{GtCO}_2/\text{year}$ and $E_{2020}^{\text{ex}} = 3.96\text{GtCO}_2/\text{year}$. Note that these values represent net emissions, i.e. after abatement. Land use emissions being exogenous, we set their path following DICE 2023 ([Barrage and Nordhaus, 2023](#)) and assume that gross emissions exogenously decline by 10% every five years and are abated at the same rate as industrial emissions.

4.2 Damages

We model production damages following DICE 2023, i.e.

$$D(Z_t) = a_1 Z_t + a_2 Z_t^{a_3} \quad (27)$$

with $a_1 = 0$ and $a_3 = 2$. This quadratic specification is identical to that of previous versions of the DICE model, but the damage level has been significantly revised: with $a_2 = 0.003467$, DICE 2023

⁴It is the approach adopted by [Benmir and Roman \(2022\)](#) who impose an emission cap but do not study the SCC. A third approach, adopted by [Barrage \(2020\)](#), is to directly calibrate the model to the world economy. In this latter case, one needs to assume that there exists a global planner choosing a carbon tax as well as global income taxes and transfers.

assumes that damages amount to 3.12% of output at 3°C warming, and 12.5% at 6°C warming. We adjust this parameter to split damages between production and utility. Following [Barrage \(2020\)](#), we assume that 74% of damages at 2.5°C warming come from output losses, and 26% come from direct utility impacts. This leads to $a_2 = 0.002566$, and it enables us to determine the parameters associated with utility damages (α_0 , see below).

4.3 Households

[To be completed].

4.4 Production

We assume that both sectors feature a Cobb-Douglas technology, and we parametrize the production functions as in [Douenne et al. \(2023\)](#), i.e.

$$F(K_{1,t}, H_{1,t}, E_t) = K_{1,t}^\alpha H_{1,t}^{1-\alpha-\nu} E_t^\nu, \quad (28)$$

$$G(K_{2,t}, H_{2,t}) = K_{2,t}^{\alpha_E} H_{2,t}^{1-\alpha_E}, \quad (29)$$

with $\alpha = 0.3$, $\nu = 0.04$ (from [Golosov et al., 2014](#)), and $\alpha_E = 0.597$ (from [Barrage, 2020](#)). We calibrate initial total factor productivities to match global GDP (from the World Bank) and aggregate industrial emissions (from [Friedlingstein et al., 2022](#)), and we borrow their growth rates from DICE 2023. Finally, we adapt the abatement cost function from DICE 2023, so that

$$\Theta(\mu_t)E_t = P_t^{\text{back}} \frac{\mu_t^{c_2}}{c_2} E_t, \quad (30)$$

with $c_2 = 2.6$, and P_t^{back} the backstop price that starts at 696.2\$/tCO₂ in 2020 and declines by 1% per year until 2050 and by 0.1% per year thereafter.

4.5 Government

[To be completed].

5 Main results

[To be completed].

6 Conclusion

[To be completed].

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Appendices

A Calibration

Table I: Calibration Summary [Work in progress].

Parameter	Description	Value	Source
<u>Preferences</u>			
β	Utility discount rate (per year)	1/(1.01)	DICE 2023
σ	Inverse of IES	1.5	DICE 2023
η^F	Frisch elasticity of labor supply	0.75	Chetty et al (2011)
ς	Labor disutility coefficient	???	To target η^F and h_{2020}
γ	Labor disutility exponent	???	To target η^F and h_{2020}
α_0	Relative preference for the environment	???	Barrage (2020) (adjusted)
<u>Production damages</u>			
a_1	Damage intercept	0	DICE 2023
a_2	Damage coefficient quadratic term	0.002566	DICE 2023 adjusted
a_3	Damage exponent	2	DICE 2023
<u>Production first sector</u>			
α	Return to scale on labor sector 1	0.3	DICE 2023
ν	Return to scale on energy sector 1	0.04	Golosov et al (2014)
δ	Depreciation rate on capital (per year)	0.1	DICE 2023
$r_{2020} - \delta$	Initial net rate of return on capital	???	At steady state
Y_{2020}	Initial output (in trillions 2023 USD)	83.476	World Bank (2016-2020)
$hh_{1,2020}$	Initial share of labor in sector 1	???	To equate MPL across sectors
$kk_{1,2020}$	Initial share of capital in sector 2	???	To equate MPL across sectors
h_{2020}	Initial labor supply per capita	0.277*	Computed from SCF
$A_{1,2020}$	Initial TFP sector 1	???	To target Y_{2020}
<u>Production second sector</u>			
α_E	Return to scale on capital sector 2	0.597	Barrage (2020)
E_{2020}	Init. gross indus. emissions (GtCO ₂ per year)	38.23	Friedlingstein et al (2022)
$A_{2,2020}$	Initial TFP sector 2	???	To target E_{2020}
<u>Climate</u>			
S_{2020}	Initial cumulative carbon emissions (in GtCO ₂)	2390	IPCC (2021)
T_{2020}	Initial atmos. temp. change (C since 1900)	1.07	IPCC (2021)
ϵ	Initial pulse-adjustment timescale	0.5	Dietz and Venmans (2019)
ζ	Trans. clim. resp. to cum. emissions (TCRE)	0.00045	IPCC (2021)

Calibration Summary [Work in progress] (continued).

E_{2020}^{land}	Init. gross CO ₂ emis. land (GtCO ₂ per year)	4.17	Friedlingstein et al (2022)
$g_{E^{\text{land}}}$	Ex. decline rate of gross land emissions (per period)	0.1	DICE 2023
<u>Abatement costs</u>			
P_{2020}^{back}	Backstop price in 2020 (in \$/tCO ₂)	696.2	DICE 2023
g_{2020}^{back}	Decline rate backstop price 2020-2050 (per year)	1%	DICE 2023
g_{2050}^{back}	Decline rate backstop price after 2050 (per year)	0.1%	DICE 2023
c_2	Exponent abatement cost function	2.6	DICE 2023
μ_{2020}	Initial abatement share	0.0513	DICE 2023
<u>Government</u>			
G_t/Y_t	Government spending to GDP ratio	0.3030*	IMF-GFS
B_{2020}	Initial net public debt to GDP ratio	0.2220*	IMF-GFS
$\tau_{H,2020}$	Initial tax rate on labor income	0.255*	Trabandt & Uhlig (2012)
$\tau_{K,2020}$	Initial tax rate on capital income	0.411*	Trabandt & Uhlig (2012)
<u>Exogenous growth parameters</u>			
$g_{A_1,2020}$	Initial TFP growth rate sector 1 (per period)	0.082	DICE 2023
$gg_{A_1,t}$	Decline rate TFP growth sector 1 (per year)	0.0072	DICE 2023
$g_{A_2,2020}$	Initial TFP growth rate sector 2 (per period)	0.082	DICE 2023
$gg_{A_2,t}$	Decline rate TFP growth sector 2 (per year)	0.0072	DICE 2023
N_{2020}	Initial population (in millions)	1,368	World Bank US-adjusted
N_{max}	Asymptotic population (in millions)	1,910	DICE 2023 US-adjusted
g_N	Rate of convergence of population	0.145	DICE 2023